

Acute Complications During Hemodialysis

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Although technical advances in hemodialysis (HD) have made the procedure increasingly safe and well tolerated, important acute complications can occur in patients receiving HD in both acute and chronic clinical settings. These complications, etiology, and their management are discussed in this chapter.

CARDIOVASCULAR COMPLICATIONS

Intradialytic Hypotension

Intradialytic hypotension occurs in 15% to 50% of treatments depending on the definition and ranges from asymptomatic episodes to marked compromise of organ perfusion, resulting in myocardial ischemia, cardiac arrhythmias, vascular thrombosis, loss of consciousness, seizures, or death.¹⁻³ Furthermore, in patients with acute kidney injury, intradialytic hypotension may reduce the likelihood of recovery. Recurrent HD-induced ischemic cardiac injury (myocardial stunning) is a prominent cause of intradialytic hypotension and is more common at higher ultrafiltration rates and less frequent with longer and more frequent dialysis treatments.⁴ Like intradialytic hypotension and postdialysis orthostatic hypotension, myocardial stunning is an independent risk factor for mortality.⁵ The pathogenesis of intradialytic hypotension is complex and is summarized in [Fig. 100.1](#). Most commonly, intradialytic hypotension results from the need to deal with excessive fluid weight gain between dialysis treatments. The subsequent rate of fluid removal required exceeds the achievable rate of intravascular filling, resulting in relative intravascular volume depletion. Recent data demonstrate an increased risk for death as the ultrafiltration rate increases, regardless of the threshold used.⁶

The immediate treatment for intradialytic hypotension is to restore the circulating blood volume by placing the patient in the Trendelenburg position, reducing or stopping ultrafiltration, and infusing boluses of 0.9% isotonic saline (≥ 100 mL, as necessary). Salt-poor albumin and other hypertonic solutions offer no proven advantage over isotonic saline and are more expensive.⁷ Blood flow rate should not be routinely reduced to manage hypotension, as this has not been shown to be beneficial and will compromise overall solute clearance. Of course, in the setting of excessive weight gain, the administration of boluses of saline or cessation of ultrafiltration makes achieving the target dry weight more difficult. Because cardiac factors can precipitate intradialytic hypotension, the clinician should maintain a high index of suspicion for cardiac ischemia, especially if hypotension is accompanied by chest pain or dyspnea, and an electrocardiogram and serum troponin values should be obtained. Similarly, recurrent and unexplained episodes of hypotension might warrant echocardiography to rule out pericarditis or pericardial effusion.

Preventive strategies include correction of anemia, treatment of congestive heart failure or arrhythmias, avoidance of antihypertensive

drugs on the day of or at least in the hours before dialysis (especially those that reduce arteriolar tone), and avoiding food before and during dialysis, as these may result in shunting of blood to the gut. Patients should be counseled to avoid excessive interdialytic weight gain (IDWG), because this is the predominant cause of intradialytic hypotension and particular attention should be given to ensuring minimal salt intake. Finally, an accurate assessment of the patient's dry weight is required. Midodrine, an oral selective $\alpha 1$ -agonist, 5 to 10 mg before dialysis, can be a useful preventive therapy if the earlier-mentioned strategies fail.

Preventive strategies through modification of the dialysis procedure include use of bicarbonate dialysate (as opposed to acetate buffered dialysate) and volumetric control of ultrafiltration. Subsequently, reducing the ultrafiltration rate by increasing either treatment time or the frequency of dialysis can be tried.^{2,8} The strategy of isolated ultrafiltration followed by HD (i.e., sequential dialysis) has been recommended for many years, although there is limited evidence to support it.² A higher dialysate sodium may improve single-treatment blood pressure but at the expense of higher IDWG and volume overload in the longer term² (see [Chapter 42](#)). The data to support sodium profiling or modeling are limited, with one study suggesting that the routine use of sodium modeling/profiling to limit or prevent intradialytic hypotension is associated with increased all-cause mortality.⁹ Online blood volume monitoring and biofeedback techniques have been developed in an attempt to improve intradialytic cardiovascular (CV) stability.¹⁰ Although online blood volume devices may decrease the incidence of intradialytic hypotension in an at-risk population, there is limited evidence that blood volume monitoring can predict intradialytic hypotension in individual patients or produce a long-term morbidity and mortality benefit, especially in the wider HD population.¹⁰ A recent small crossover study failed to show a benefit of ultrafiltration profiling in preventing intradialytic hypotension or treatment-related cardiac stress.¹¹ Cooling of dialysate to 35.5°C to 36°C (95.9–96.8°F), a measure that induces release of catecholamines, resulting in vasoconstriction or at least preventing vasodilation, may lessen hypotension. Modulation of dialysate temperature is achieved by measuring the blood temperature in the arterial and venous circuits and feeding back the information to the arterial and venous thermostats in the dialysis machine. The machine can be programmed to allow a constant body temperature and a negative overall energy transfer, so-called *isothermic* or *thermoneutral* HD, which aims to prevent energy transfer between the dialysate and extracorporeal blood. A recent systematic review of the randomized controlled trials using biofeedback-modulating dialysate temperature demonstrated significant reductions in the incidence of intradialytic hypotension compared with conventional dialysis.¹² However, large multicenter randomized trials are needed to determine whether reduced temperature dialysis reduces CV events and patient death.

Causes of Intradialytic Hypotension

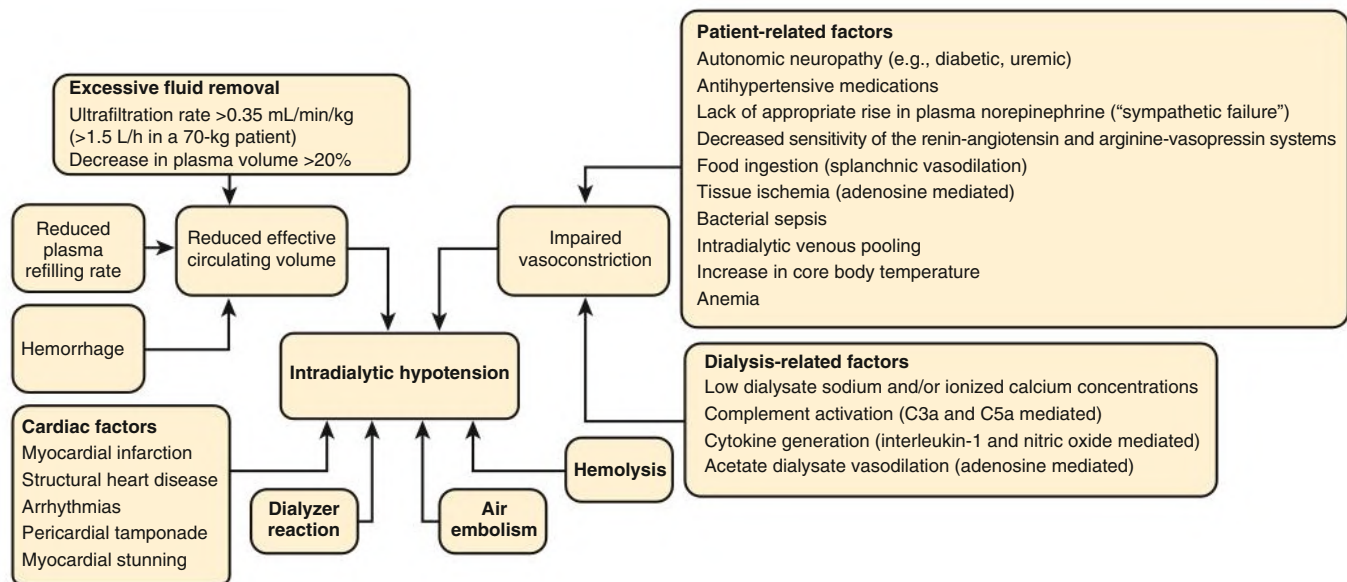


Fig. 100.1 Pathogenesis and causes of intradialytic hypotension.

Across Europe and many other countries, hemodiafiltration (HDF) has become a common mode of dialytic therapy. Although claims of improved mortality with this modality remain a matter of debate, most accept that HDF is associated with improved intradialytic CV stability and less frequent hypotension, but this is also disputed by some.¹³

Intradialytic Hypertension

Intradialytic hypertension (IDH) occurs in 5% to 30% of treatments depending on the definition used.^{2,14} Hypertension during or immediately after HD constitutes an important risk factor for CV mortality. Moreover, an intradialytic increase in systolic blood pressure (BP) is associated with an increased risk for hospitalization or death.¹⁵

In most circumstances, an intradialytic elevation of BP indicates significant volume overload. However, in a number of patients BP remains elevated despite fluid removal, a syndrome called *dialysis-refractory hypertension*. These patients are usually young, with pre-existing hypertension, and have excessive IDWG and a hyperactive renin-angiotensin system in response to fluid removal.¹⁶ In these patients the hypertension is still mediated by the previous volume expansion, but there is often a lag of days to 2 or more weeks before the BP becomes normal after reduction of the dry target weight (known as the dialysis "lag phenomenon"). Often patients can discontinue most of their antihypertensive agents once this period has elapsed. Chapter 42 presents more details on the diagnosis and management of hypertension associated with hemodialysis.

Erythropoietin (EPO) and other erythropoiesis-stimulating agents have been associated with a 20% to 30% incidence of new-onset hypertension or exacerbation of hypertension. Furthermore, among patients receiving intravenous (not subcutaneous) EPO, elevated levels of endothelin-1 (a potent vasoconstrictor) have been shown to correlate with increased BP. Such EPO-induced rises in BP are usually associated with a rapid rise in hemoglobin and can be avoided by a more conservative approach to correcting the hemoglobin.

IDH can be precipitated by the use of high-sodium dialysate, which is intended to mitigate the intradialytic decrease in serum osmolality

that occurs with the diffusive removal of urea and sodium.¹⁷ Although this approach stabilizes BP during dialysis by improving intravascular filling, high-sodium dialysate results in a positive intradialytic sodium balance and is associated with increased postdialysis thirst, resulting in significant weight gain in the interdialytic period. To circumvent these problems, sodium modeling has been adopted as an approach that uses variable sodium concentrations in the dialysate, generally with sodium reduced in a continuous or stepwise manner from an initial level of 150 to 154 mmol/L to 138 to 142 mmol/L. Although sodium modeling has been widely promoted, results from both randomized and non-randomized crossover studies do not suggest any definitive benefit.^{9,18} Other hypothesized mechanisms of IDH include hyperactivity of the sympathetic nervous system¹⁹ and increased cardiac output resulting from fluid removal, particularly in patients with cardiomyopathy.²⁰ Clinicians also should be aware of possible dialytic removal of certain antihypertensive drugs, such as angiotensin-converting enzyme (ACE) inhibitors and β -blockers.

Increasing hypertension during a dialysis session requires intervention if systolic BP is greater than 180 mm Hg. This is best treated with a short-acting agent, such as the ACE inhibitor captopril. Successful treatment of hypertension for a longer period requires an accurate determination of the patient's dry weight and its achievement by gradual ultrafiltration over several weeks of dialysis. Once dry weight is achieved, optimization of antihypertensive drug therapy is warranted, potentially including the use of minimally dialyzable or nondialyzable medications such as angiotensin receptor blockers, calcium channel blockers, clonidine, and carvedilol. However, use of antihypertensive agents may make achievement of dry weight difficult because of induced intradialytic hypotension. Evidence from the Dry-Weight Reduction in Hypertensive Hemodialysis Patients (DRIP) trial²¹ suggests that optimal control of BP in HD patients is via control of extracellular fluid volume (salt and water) and not the use of antihypertensive agents. One common error in management is to treat dialysis hypertension with increasing BP medications as opposed to opting to achieve dry weight. The use of vasodilator drugs (hydralazine, minoxidil) can lead to increased fluid retention that worsens volume overload. Minoxidil

also can cause pleural and pericardial effusions and should be avoided in dialysis patients if at all possible.

Cardiac Arrhythmias

Intradialytic arrhythmias are common and are often multifactorial in origin.^{22,23} Left ventricular hypertrophy, congestive cardiomyopathy, uremic pericarditis, silent myocardial ischemia, and conduction system calcification are frequently encountered in adult dialysis patients. In addition, polypharmacy coupled with the constant alterations in fluid, electrolyte, and acid-base homeostasis may precipitate intradialytic arrhythmias. Intradialytic hypotension is also associated with intradialytic cardiac arrhythmias.²⁴ The range of electrocardiographic abnormalities that may be encountered in kidney failure is shown in Table 100.1. QTc dispersion, the difference between maximum and minimum QTc interval on a standard 12-lead electrocardiogram, is prolonged after HD and has been proposed as a prognostic indicator of cardiac complications in dialysis patients.

Preventive measures include the use of bicarbonate dialysate and careful attention to dialysate potassium and calcium levels as well as careful attention to preventing intradialytic hypotension (see earlier). There are very few trials of the most appropriate dialysate potassium level, although recent observational evidence suggests that dialysate potassium levels below 2 mmol/L should be avoided in most patients.²⁵ Zero potassium dialysate should be avoided in the hyperkalemic individual because of its arrhythmogenic potential, particularly in patients receiving digoxin. Serum digoxin levels should be regularly monitored and the need for the drug regularly reassessed.

Sudden Death

Cardiac arrest occurs in 7 per 100,000 HD sessions and is more common in the elderly, patients with diabetes, and patients using central venous catheters.²⁶ Sudden deaths during dialysis are observed more frequently after the 3-day (classically weekend) interdialytic interval in patients receiving dialysis three times per week.^{27,28} The etiology is complex but likely related to cardiac structural abnormalities related to long-standing hypertension and uremia coupled with the more marked fluid and solute accumulation seen in the long interdialytic period. Both peritoneal dialysis patients and patients undergoing frequent long-hours HD do not show this high event rate on a particular day of the week.²⁸ Recent analysis of data from implantable cardiac monitors in HD patients with sudden cardiac death found that the vast majority were due to bradycardia and asystole, rather than malignant ventricular arrhythmias, raising uncertainty regarding β -blocker use in this population.²⁹

Although coronary heart disease increases the risk for sudden death, other catastrophic intradialytic events need to be ruled out. The prompt recognition and treatment of hyperkalemia, as a reversible cause of cardiac dysfunction, is imperative. Profound generalized muscle weakness may be a warning sign of imminent life-threatening hyperkalemia.

When cardiopulmonary arrest occurs during dialysis, an immediate decision must be made as to whether the collapse is the result of an intrinsic disease or technical errors, such as air embolism, unsafe dialysate composition, overheated dialysate, line disconnection, or sterilant in the dialyzer. Air in the dialysate, grossly hemolyzed blood, and hemorrhage caused by line disconnection can be easily detected. However, if no obvious cause is identifiable, blood should not be returned to the patient, particularly if the arrest occurred immediately on initiation of dialysis. A patient exposed to formaldehyde may have reported earlier burning at the access site; fortunately, this agent is rarely used today. If the possibility of a problem with dialysate composition is remote, blood may be returned to the patient. However, blood and dialysate samples should be immediately sent for electrolyte analysis, the dialyzer and

TABLE 100.1 Electrocardiographic Abnormalities in Kidney Failure

Function	Abnormality
PR interval	Usually normal; prolongation in long-term HD. Calcification of mitral valve annulus may involve His bundle, resulting in complete heart block.
QRS interval	
Amplitude	Increases during ultrafiltration (correlates with reduction in LV dimensions). LVH on voltage criteria found in up to 50%.
Duration	Prolonged (within normal range) by hemodialysis. Late potentials increased only in patients with preexisting coronary heart disease. Prolonged in hyperkalemia.
ST segment	Depression during HD does not predict coronary artery disease. Depression or elevation may occur in hyperkalemia. Depression during ambulatory monitoring poorly predictive of coronary artery disease.
QTc interval	Increases during HD (correlates with reduction in K^+ and Mg^{2+}). Increased QT dispersion reported in patients on dialysis.
T wave	Peaking or inversion may occur in hyperkalemia. Inversion in anterolateral leads in LVH with strain pattern.
Rhythm	Bradycardia and asystole in long intradialytic break and during HD. Atrial and ventricular arrhythmias during HD.

Risk factors include LV dysfunction, wall motion abnormalities, known coronary artery disease, abnormal perfusion scans (even without coronary artery disease), use of cardiac glycosides, and low dialysate potassium concentration.

HD, Hemodialysis; LV, left ventricle; LVH, left ventricular hypertrophy.

blood lines saved for later analysis, and the dialysis machine replaced until all its safety features have been thoroughly evaluated for possible malfunction. It should be standard practice to have defibrillators in dialysis units. The management of cardiopulmonary arrest during dialysis should follow the standard principles of cardiopulmonary resuscitation; the diagnosis and management of technical errors are discussed later. All dialysis units should have established protocols for managing cardiac arrest and other common dialysis emergencies.

Prevention of sudden cardiac death in HD patients, including the role of implantable defibrillators, is discussed further in Chapter 85.

Dialysis-Associated Steal Syndrome

The clinical presentation, differential diagnosis, and evaluation of dialysis-associated steal syndrome are discussed further in Chapter 96.

NEUROMUSCULAR COMPLICATIONS

Muscle Cramps

Muscle cramps occur in 5% to 20% of patients late during dialysis and frequently involve the legs. They account for 15% of premature discontinuations of dialysis.³⁰ Electromyography shows increased tonic muscle electrical activity throughout dialysis, and serum creatine kinase may be elevated.

Although the pathogenesis is unknown, dialysis-induced volume contraction and hypoosmolality appear to be common predisposing factors. Indeed, the onset of muscle cramps may give an indication that

the target weight has been reached. However, hypomagnesemia and carnitine deficiency also may play a role.

The acute management is directed at increasing plasma osmolality. Cessation of ultrafiltration is not useful. Parenteral infusion of 23.5% hypertonic saline (15–20 mL), 25% mannitol (50–100 mL), or 50% dextrose in water (25–50 mL) is equally effective. However, hypertonic saline may result in postdialytic thirst, and both hypertonic saline and mannitol cause transient warmth and flushing during the infusion. Furthermore, large and repetitive infusions of mannitol may lead to increased thirst, IDWG, and fluid overload. Overall, dextrose in water is preferred, particularly in nondiabetic patients.

Preventive measures include dietary counseling about excessive IDWG, in particular reducing dietary sodium intake. In patients without clinical signs of fluid overload, it is reasonable to increase the dry weight by 0.5 kg and to observe the clinical response. Quinine sulfate (250–300 mg) or oxazepam (5–10 mg) given 2 hours before dialysis also may be effective. Although the US Food and Drug Administration regards quinine sulfate as unsafe and ineffective for the prevention of cramps, this drug works well in some patients and is used freely in most parts of the world. Some reports also promote the use of vitamin E in this role.³¹ The use of sodium gradients during dialysis may have some benefit as well. Proposed strategies include starting with a dialysate sodium concentration of 145 to 155 mmol/L and a linear decrease to 135 to 140 mmol/L by the completion of the treatment. A comparison of sodium modeling with an exponential, linear, or step program has yielded similar results.^{18,32} In anecdotal reports, 5 mg of enalapril twice weekly may be effective, presumably by inhibiting angiotensin II-mediated thirst. Stretching exercises, magnesium, creatine monohydrate (12 mg before dialysis), and L-carnitine supplementation (20 mg/kg per dialysis session) also may be beneficial. An intradialytic blood volume biofeedback control system has been shown to reduce the incidence of muscle cramps.³³

Restless Legs Syndrome

Restless legs syndrome is common in dialysis patients. The typical report is of crawling sensations in the legs that occur with inactivity, and symptoms may worsen during dialysis. The etiology, prevention, and management of restless legs syndrome are discussed in [Chapter 89](#).

Dialysis Disequilibrium Syndrome

Despite a decline in its incidence, dialysis disequilibrium syndrome (DDS) is still observed sporadically in patients who are initiated on HD on high-flux dialyzers with large surface areas and short dialysis time. Risk factors ([Table 100.2](#)) include young age, severe uremia, rapid and marked intradialytic falls in urea at dialysis initiation, low dialysate sodium concentration, and preexisting neurologic disorders (see [Chapter 89](#)).

DDS commonly manifests with restlessness, headache, nausea, vomiting, blurred vision, muscle twitching, disorientation, tremor, and hypertension. More severe manifestations include obtundation, seizures, and coma. DDS usually develops toward the end of dialysis but may be delayed for up to 24 hours. Although cerebral edema is a consistent finding on computed tomographic scanning, DDS remains a clinical diagnosis because laboratory tests, including electroencephalography, are nonspecific. It is usually self-limited, but full recovery may take several days. The differential diagnoses for DDS are detailed in [Table 100.2](#).³⁴

The pathogenesis of DDS is still a subject of debate. The reverse urea effect theory, which proposes that a transient osmotic disequilibrium occurs during dialysis as a result of a more rapid removal of urea from blood than from cerebrospinal fluid, has been disputed.³⁵ In animals undergoing rapid dialysis, despite the correction of

TABLE 100.2 Dialysis Dysequilibrium

Differential diagnosis	Subdural hematoma Uremia Nonketotic hyperosmolar coma Acute cerebrovascular event Dialysis dementia Excessive ultrafiltration and seizure Hypoglycemia Malignant hypertension Hyponatremia
Risk factors	Younger age Severe uremia Rapid and marked intradialytic fall in urea at dialysis initiation Low dialysate sodium concentration Preexisting neurologic disorders
Prevention	Early recognition of uremia states Stepped initiation of dialysis with short initial treatment times Use of volumetric-controlled hemodialysis machines Low blood flow rates Utilize small surface area dialyzers Targeted reduction in blood urea ~30% Sodium profiling, high not low sodium dialysate concentration

systemic acidosis, a paradoxical cerebrospinal fluid acidosis develops that is aborted by slower dialysis. An additional mechanism is the intracerebral accumulation of idiogenic osmoles, such as inositol, glutamine, and glutamate.

In high-risk patients (see [Table 100.2](#)), preventive measures include the use of volumetric-controlled machines, bicarbonate dialysate, sodium modeling, earlier recognition of uremic states, and stepped initiation of dialysis (short initial treatment times with lower blood pump speeds). To this end, short and more frequent dialysis treatments are recommended with use of small surface area dialyzers and reduced blood flow rates. The target reduction in blood urea should initially be limited to 30%. The prophylactic use of mannitol or anti-convulsants is not recommended.

An extension of this syndrome may be one that mimics osmotic demyelination syndrome, similar to that seen with rapid correction of hyponatremia. Several cases have been reported in association with dialysis initiation, with clinical manifestations similar to those of the locked-in pontine picture of central demyelination. The difference is that with the dialysis-related condition, patients appear to recover over the ensuing 5 to 7 days, and the condition seems to be related to edema rather than demyelination.³⁶

Seizures

Intradialytic seizures occur in fewer than 10% of patients and tend to be generalized but easily controlled. However, focal or refractory seizures warrant evaluation for focal neurologic disease, particularly intracranial hemorrhage. Causes of seizures are summarized in [Fig. 100.2](#) and are discussed further in [Chapter 89](#).

Treatment of established seizures requires cessation of dialysis, maintenance of airway patency, and investigation for metabolic abnormalities. Intravenous diazepam, alprazolam, or clonazepam, and phenytoin may be required. Intravenous 50% dextrose in water should be administered promptly if hypoglycemia is suspected.

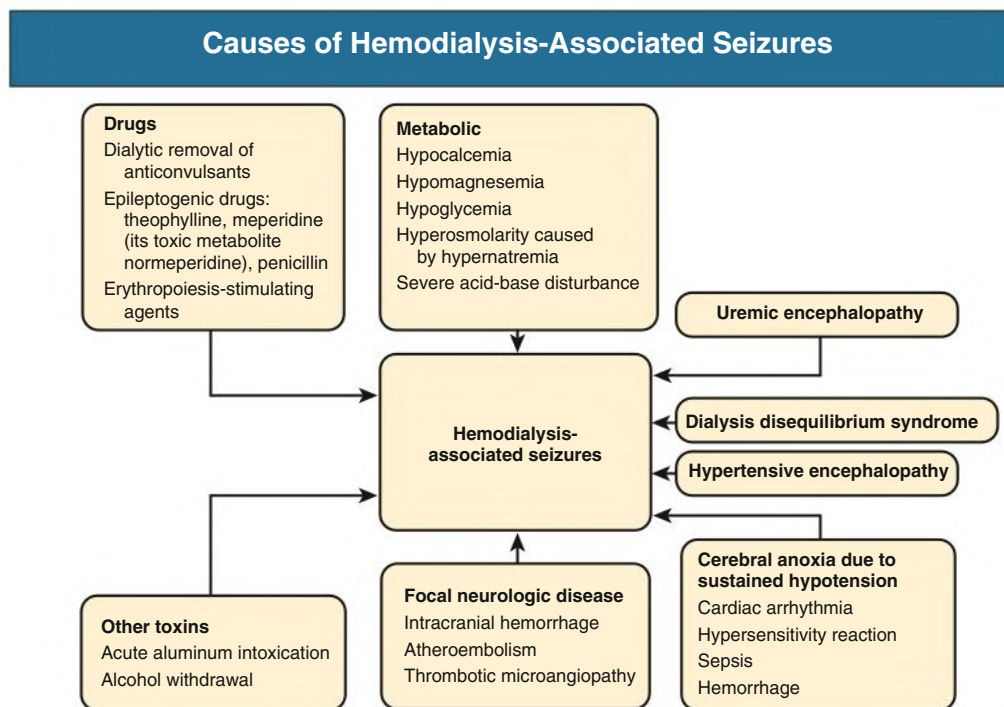


Fig. 100.2 Causes of hemodialysis-associated seizures.

Headache

Dialysis headache is common and typically consists of a bifrontal discomfort that develops during dialysis and may become intense and throbbing, accompanied by nausea and vomiting. It is usually aggravated by the supine position, but there are no visual disturbances.³⁷

Although its cause is unknown, dialysis headache may be a subtle manifestation of DDS or may be related to accumulation of acetate, which is present in low concentrations (3–4 mmol/L) in almost all dialysate fluid. A role for nitric oxide also has been postulated. Alternatively, it may be a manifestation of caffeine withdrawal caused by dialytic removal of caffeine.

Management consists of oral analgesics (e.g., acetaminophen [paracetamol]). Preventive measures include slow dialysis with reduced blood flow rates; change to bicarbonate dialysate, sodium, and ultrafiltration modeling; coffee ingestion during dialysis; and use of reprocessed dialyzers. Attention should also be paid to adequate flushing of the lines and dialyzer before commencement of dialysis. Changing the dialyzer (especially to a synthetic membrane or an alternative synthetic membrane) is sometimes useful in refractory cases.

HEMATOLOGIC COMPLICATIONS

Complement Activation and Dialysis-Associated Neutropenia

Complement activation and neutropenia, first seen with unsubstituted cellulose dialyzers that are now not used, can occur with widely used dialyzer membranes, including cellulose acetate and “synthetic” membranes such as polysulfone. However, this is very uncommon, especially in the case of the synthetic membranes, and the long-term clinical relevance of this phenomenon remains speculative. Its contribution to acute intradialytic morbidity is discussed later.

Intradialytic Hemolysis

Acute hemolysis can be caused by faulty dialysis equipment, chemicals, drugs, toxins, or patient-related factors (Fig. 100.3).³⁸ With the advent

of better dialysis equipment design and the widespread use of reverse osmosis and/or deionization systems and carbon filters, traumatic red blood cell (RBC) fragmentation caused by poorly designed blood pumps and methemoglobinemia caused by water contamination with chloramine or copper are rarely seen today. However, nitrate or nitrite intoxication causing methemoglobinemia still occurs sporadically in patients on home HD who use well water contaminated with urine from domesticated animals. Furthermore, during dialyzer reprocessing, formaldehyde retention can result in hemolysis by inducing formation of cold agglutinins or inhibiting RBC metabolism.

The diagnosis of acute hemolysis is evident when grossly translucent hemolyzed blood is observed in the tubing. Patients with methemoglobinemia have nausea, vomiting, hypotension, and cyanosis, and oxygen therapy does not improve the black blood present in the extracorporeal circuit. Copper contamination should be suspected in the presence of skin flushing and abdominal pain or diarrhea.

Evaluation should include reticulocyte count, haptoglobin, lactate dehydrogenase, blood smear, Coombs test, and measurement of methemoglobin. Chromium 51 (⁵¹Cr)-labeled RBC survival and bone marrow examination may occasionally be indicated if there is recurrent hemolysis. More important, analysis of tap water for chloramines and assessment of the status of charcoal filtration (which removes chloramines) and metal contaminants and thorough analysis of the dialysis equipment for clues of increased blood turbulence are recommended.

Hemorrhage

Bleeding complications are commonly related to the use of intradialytic anticoagulation, which exacerbates the uremic bleeding diathesis (see Chapter 87). In addition, dialysis patients are prone to spontaneous bleeding at specific sites, such as gastrointestinal arteriovenous malformations; subdural, pericardial, pleural, retroperitoneal, and hepatic subcapsular spaces; and the ocular anterior chamber. Despite its limitations, the bleeding time remains the best indicator of hemorrhagic tendency.

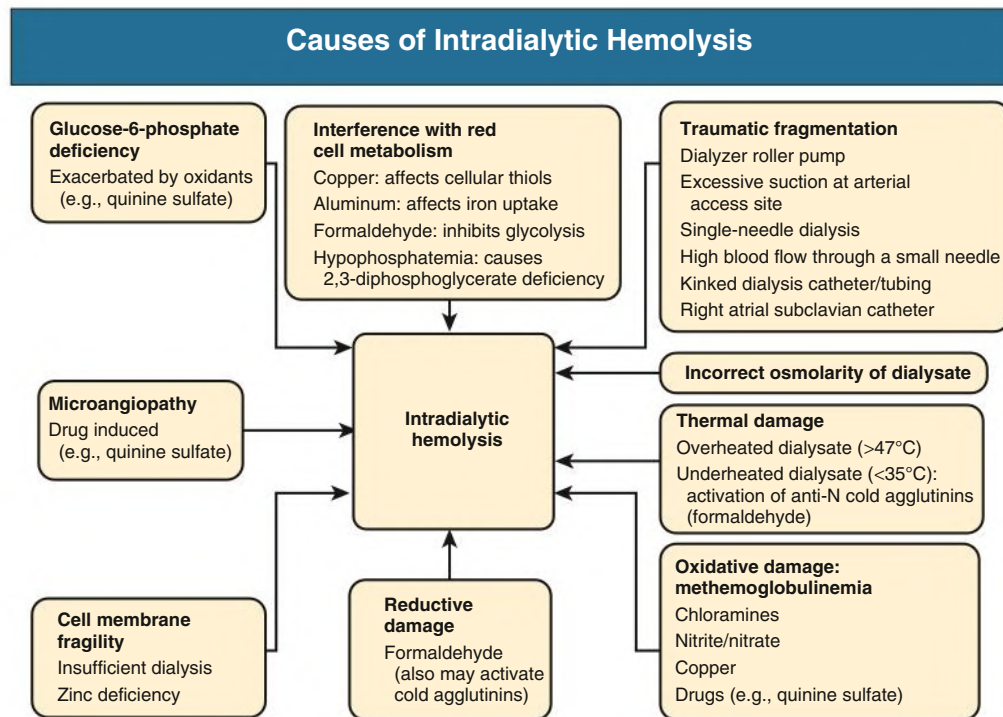


Fig. 100.3 Causes of intradialytic hemolysis.

In addition to specific measures directed to the site of hemorrhage, reversal of uremic platelet dysfunction is imperative. Strategies include the use of erythropoiesis-stimulating agents or RBC transfusions to achieve a hematocrit above 30% to improve rheologic platelet–vessel wall interactions, intravenous conjugated estrogens at 0.6 mg/kg/day for 5 consecutive days, intravenous or subcutaneous 1-deamino-8-D-arginine-vasopressin (DDAVP) at 0.3 µg/kg administered over 15 to 30 minutes, and intravenous infusion of cryoprecipitate (see Chapter 87). For patients experiencing severe bleeding, it is advisable to consider heparin-free dialysis with use of normal saline flushes every 15 to 30 minutes with ultrafiltration adjustments. An alternative approach is to utilize predilution HDF, which provides dialysate infusion prior to entry of the blood into the dialyzer. However, this approach requires high ultrafiltration volumes and does not have a proven benefit over normal saline flushes.^{39,40} Other alternatives may include regional heparin or citrate anticoagulation, and in the long term the use of low molecular weight heparin, heparin modeling, or prostacyclin may be considered. More recently, the use of heparin-bound dialyzers (such as hemophan or AN69ST)⁴¹ or citrate-containing dialysate have been advocated in patients at risk for bleeding, although they may not decrease the need for routine circuit anticoagulation.⁴² In patients scheduled for elective surgery or invasive procedures, cessation of aspirin should be considered a week in advance, the dose of anticoagulant reduced to a minimum, and the hematocrit maintained above 30%. In some patients, intravenous DDAVP to reverse the uremic platelet defect also may be required, although its efficacy has been questioned. Tranexamic acid, a potent fibrinolytic inhibitor, can be used as an adjuvant treatment to control hemorrhage in dialysis patients.⁴³

Thrombocytopenia

Heparin-induced thrombocytopenia is an increasingly important cause of thrombocytopenia in dialysis patients. The diagnosis and management, including alternative strategies for anticoagulation for HD, are discussed in Chapter 87.

PULMONARY COMPLICATIONS

Dialysis-Associated Hypoxemia

In most patients, the arterial Pao₂ decreases by 5 to 20 mm Hg (0.6–4.0 kPa) during dialysis, reaching a nadir at 30 to 60 minutes, and resolves within 60 to 120 minutes after discontinuation of dialysis. This decrease is usually of no clinical significance to patients unless there is preexisting chronic cardiopulmonary disease.

Hypoventilation is the main implicated factor and is primarily central in origin as a result of a decrease in carbon dioxide production after acetate metabolism (specific to acetate-containing dialysate), loss of carbon dioxide in the dialyzer (with both acetate and bicarbonate dialysate), and rapid alkalization of body fluids (particularly with large surface area dialyzers).⁴⁴ In addition, acetate-induced respiratory muscle fatigue can lead to hypoventilation, especially in acutely ill patients. Furthermore, ventilation-perfusion mismatch may be caused by pulmonary leukocyte agglutination (in part resulting from complement activation) or impaired cardiac output (e.g., from myocardial stunning).

In high-risk patients with fluid overload, preventive measures consist of using intradialytic oxygen supplementation, biocompatible membranes, and avoiding acetate dialysate. Optimizing hematocrit/hemoglobin values and performing sequential ultrafiltration followed by HD may further reduce the likelihood of hypoxemia.

TECHNICAL MALFUNCTIONS

Air Embolism

The most vulnerable source of air entry into the extracorporeal circuit is the prepump tubing segment, in which significant subatmospheric pressures prevail. However, other sources need to be considered, including intravenous infusion circuits especially with glass bottles, air bubbles from the dialysate, and dialysis catheters. High blood flow rates may allow rapid entry of large volumes of air despite small leaks.

Clinical manifestations depend on the volume of air introduced, the site of introduction, the patient's position, and the speed at which air is introduced.⁴⁵ In the sitting position, air entry through a peripheral vein bypasses the heart and causes venous emboli in the cerebral circulation. The acute onset of seizures and coma in the absence of precedent symptoms such as chest pain and dyspnea is highly suggestive of air embolism. In the supine position, air introduced through a central venous line will be trapped in the right ventricle, where it forms foam, interferes with cardiac output, and, if it is large enough, leads to obstructive shock. Dissemination of micro-emboli to the pulmonary vasculature results in dyspnea, dry cough, chest tightness, or respiratory arrest. Furthermore, passage of air across the pulmonary capillary bed can lead to cerebral or coronary artery embolism. In the left Trendelenburg position, air emboli migrate to the lower extremity venous circulation, resulting in ischemia as a result of increased outflow resistance. Foam may be visible in the extracorporeal tubing, and cardiac auscultation may reveal a peculiar churning sound, the so-called "millwheel murmur."

The immediate management of clinically suspected air embolism is summarized in Fig. 100.4. Prevention depends primarily on dialysis machines equipped with venous air bubble traps and foam detectors located just distal to the dialyzer and a venous pressure monitor at the venous end. The detector is attached to a relay switch that simultaneously activates an alarm, shuts off the blood pump, and clamps the venous blood line if air is detected. Therefore, dialysis should never be performed in the presence of an inoperative air detection alarm system. Glass bottles should be avoided because they create vacuum effects that can permit air entry into the extracorporeal system. Dialysis catheters should be aspirated and flushed with saline before connection. Extra care should be taken to ensure occlusion of catheters at two points at the end of treatment (clamp and line

cap). Dialyzer rinsing, before use, should expand all compartments to remove residual air bubbles.

Incorrect Dialysate Composition

Incorrect dialysate composition results from technical or human errors. Because the primary solutes constituting the dialysate are electrolytes, the dialysate concentration will be reflected by its electrical conductivity. Therefore, proper proportioning of concentrate to water can be achieved by the use of a meter that continuously measures the conductivity of the dialysate solution as it is being fed to the dialyzer. Life-threatening electrolyte and acid-base abnormalities are avoidable if the conductivity alarm is functioning properly and the alarm limits are set correctly. However, in dialysis machines that are equipped with conductivity-controlled mixing systems, the system automatically changes the mixing ratio of the concentrates until the dialysate solution conductivity falls within the set limits. This may inadvertently lead to dialysate without any bicarbonate, with apparently acceptable conductivity. Therefore, if conductivity-controlled systems are used, it is safer to also check the dialysate pH before dialysis. Conductivity monitors can fail or can be improperly adjusted due to human error. Therefore, it is important to add human monitoring of dialysate composition before every treatment, whenever a machine has been sterilized or moved about, or whenever a new concentrate is used. Furthermore, as many nonstandard solutions are available, some of which may be used with an inappropriate proportioning system, it is also essential that the supplies match the machine-proportioning ratio for which they were prepared for the appropriate final dialysate composition to be obtained (note that different machine manufacturers use different proportioning ratios). Double checking of concentrate variations (e.g., potassium and calcium concentration) should be practiced, as it is for intravenous drug administration.

Hypernatremia

Intradialytic hypernatremia occurs when concentrate or the ratio of concentrate to water is incorrect and the conductivity monitors or the alarms are not functioning properly. Hyperosmolality results in intracellular water depletion. Clinical manifestations include thirst, headache, nausea, vomiting, seizures, coma, and death. Aggressive treatment is mandatory and includes cessation of dialysis, hospitalization, and infusion of 5% dextrose in water. Dialysis should be resumed with a different machine; the dialysate sodium level should be 2 mmol/L lower than the plasma level, and isotonic saline should be concurrently infused. Dialysis against a sodium level 3 to 5 mmol/L lower than the serum level may increase the risk for disequilibrium. Ultrafiltration with equal volume replacement with normal saline is another option.

Hyponatremia

Failure to add concentrate, inadequate concentrate-to-water ratio, or conductivity monitor or alarm malfunction can cause intradialytic hyponatremia. Hyponatremia also can occur during the course of dialysis with a proportioning system if the concentrate container runs dry and the conductivity set limits are inappropriate. Acute hypoosmolality causes hemolysis with hyperkalemia and hemodilution of all plasma constituents. Symptoms include restlessness, anxiety, pain in the vein injected with the hypotonic hemolyzed blood, chest pain, headache, nausea, and occasional severe abdominal or lumbar cramps. Pallor, vomiting, and seizures may be observed. Treatment consists of clamping the blood lines and discarding the hemolyzed blood in the extracorporeal circuit. High-flow oxygen and cardiac monitoring are imperative because of hyperkalemia and potential myocardial injury. Dialysis should be restarted with a new dialysate batch containing

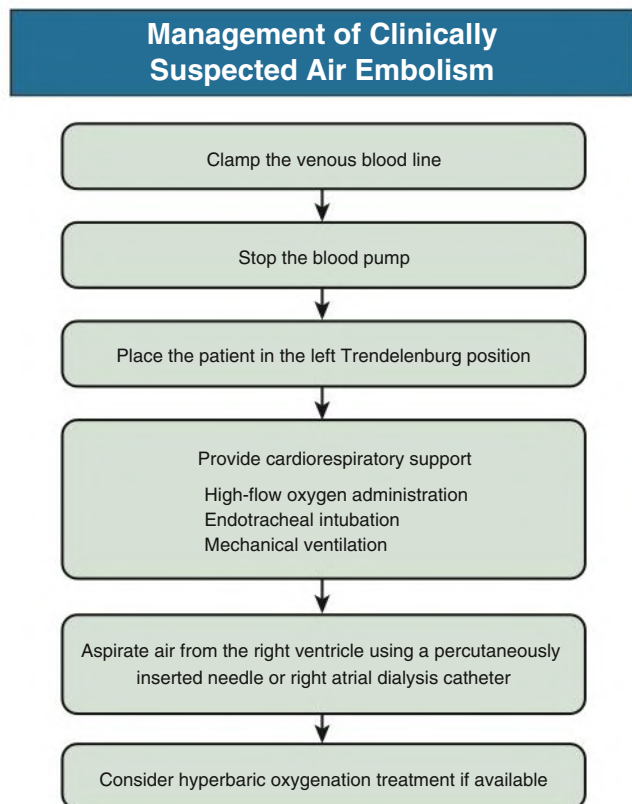


Fig. 100.4 Management of clinically suspected air embolism.

low potassium, and high transmembrane pressure should be applied to remove excess water. Correction of serum sodium concentration should be achieved by no more than 1 to 2 mmol/L/h. Anticonvulsants are indicated for seizures and blood transfusions for severe anemia. Successful correction of severe hyponatremia has been reported in a single 3-hour HD session with a dialysate sodium concentration of 135 mmol/L without any adverse neurologic consequences despite a serum sodium correction rate of 3 mmol/L/h.⁴⁶ This suggests that elevated blood urea levels might protect uremic patients from the development of demyelinating syndromes when hyponatremia is rapidly corrected.

Metabolic Acidosis

Although acute intradialytic metabolic acidosis can be a manifestation of improper mixing of concentrates or failure of pH monitors, other causes need to be ruled out, including diabetic or alcoholic ketoacidosis, lactic acidosis, ischemic tissues, toxic ingestions, and dilutional acidosis.⁴⁷ The diagnosis is usually suggested by the acute onset of hyperventilation during HD and confirmed by laboratory evaluation. In most circumstances, correction of the underlying cause and use of bicarbonate dialysate at the appropriate concentration (32–35 mmol/L) are adequate measures.

Metabolic Alkalosis

Severe intradialytic metabolic alkalosis is rare and may be caused by errors in dialysate concentrates, reversed connection of bicarbonate and acid concentrate containers to the entry ports of the dialysis machine, pH monitor malfunction, or use of regional citrate anticoagulation. The most common cause, however, is hydrochloric acid loss as a result of vomiting or nasogastric suction. Attention should also be directed to identification of sources of added alkali.⁴⁸

Acute treatment is rarely necessary unless a technical error has occurred. Removal of the alkali source is usually sufficient, and H₂-antagonists or proton pump inhibitors may be successful if there is gastric acid loss. The administration of sodium chloride to anephric patients with chloride-sensitive alkalosis will not repair the alkalosis. If a more rapid reduction in serum bicarbonate is desired, modification of the dialysate bath by replacement of alkali with chloride, substitution of bicarbonate with acetate dialysate, use of acid dialysate, and infusion of hydrochloric acid are effective but cumbersome measures. The use of conventional or low-bicarbonate (25–30 mmol/L) dialysate is probably as effective.

Temperature Monitor Malfunction

Malfunction of the thermostat in the dialysis machine can result in the production of excessively cool or hot dialysate. Whereas cool dialysate is not dangerous and may have beneficial hemodynamic effects (see discussion regarding intradialytic hypotension), overheated dialysate can cause immediate hemolysis and life-threatening hyperkalemia, particularly if the dialysate temperature increases to more than 51°C. In such an event, dialysis must be stopped immediately and blood in the system discarded. The patient should be monitored for hemolysis and hyperkalemia. Dialysis should be resumed to cool the patient by use of a dialysate temperature of 34°C to treat hyperkalemia and allow blood transfusions if necessary. Visual and audible alarms are mandatory to prevent this complication.

Blood Loss

Intradialytic blood loss can result from arterial or venous needle disengagement from the access, separation of the venous or arterial line connections, femoral or central line dialysis catheter perforation or dislodgment, or rupture of a dialysis membrane with or without malfunction of the blood leak detector. Care should be taken to ensure

needle insertion sites and line connection points are and remain visible during a dialysis treatment. Clinical findings include hypotension, loss of consciousness, and cardiac arrest. In addition, after traumatic insertion of a dialysis catheter, blood loss can result in pain or mass from a rapidly expanding hematoma; chest, shoulder, or neck pain from intrapericardial blood loss; back, flank, groin, or lower abdominal pain or distention from retroperitoneal bleeding; or hemoptysis from pulmonary bleeding. Acute management includes the discontinuation of HD, pressure application for local hemostasis, reversal of anticoagulation (e.g., protamine sulfate for heparin), hemodynamic support, oxygen administration, and surgical intervention if needed.

In the home dialysis setting, especially nocturnal HD, venous needle dislodgement is potentially catastrophic. A number of devices have been developed for early detection of blood leak resulting from needle dislodgment, including bed-wetting incontinence pads, Red-Sense and HEMODialert blood detectors, and other proprietary techniques. However, the most important component is prevention by comprehensive and repeated patient education about adequate taping of dialysis needles and lines.

Clotting of Dialysis Circuit

Clotting of the extracorporeal circuit during dialysis is a common practical problem, has many underlying causes, and warrants a thorough investigation. Technical factors include an inadequate or poor priming technique, resulting in retention of air in the dialyzer, and lack of or inadequate priming of the heparin infusion line. Such operator-induced errors are corrected through ongoing staff education and competency assessment. Incorrect heparin loading dose, insufficient time lapse after loading dose of heparin for systemic anticoagulation to occur, incorrect pump setting for constant heparin infusion, delayed start of the heparin pump, and failure to release the heparin line clamp are important correctable causes of clotting that also should be considered. Low molecular weight heparin is often used as a single bolus at the commencement of dialysis but may not maintain anticoagulation for extended hours of treatment (e.g., nocturnal HD). Vascular access-related problems from inadequate blood flow caused by needle or catheter positioning or clotting, excessive access recirculation, and frequent interruption of blood flow also can result in clotting. Immediate management requires prompt recognition of the underlying cause and implementation of corrective actions, including ongoing heparin dose adjustment and, if indicated, vascular access revision.

Heparin-free dialysis is sometimes medically indicated, such as in an actively bleeding patient or postoperatively. Although traditional approaches to avoid circuit clotting in this circumstance involve frequent saline flushes, an alternative approach is to employ predilution HDF as outlined earlier.

DIALYSIS REACTIONS

During HD, blood is exposed to surface components of the extracorporeal circuit, including the dialyzer, tubing, sterilization processes, and other foreign substances related to the manufacturing and reprocessing procedures. This interaction between the patient's blood and the extracorporeal system can lead to various adverse reactions (Fig. 100.5).

Anaphylactic and Anaphylactoid Reactions

Clinical Presentation

Anaphylaxis is the result of an immunoglobulin E (IgE)-mediated acute allergic reaction in a sensitized patient, whereas anaphylactoid reactions result from the direct release of mediators by host cells. Symptoms usually develop within the first 5 minutes of dialysis,

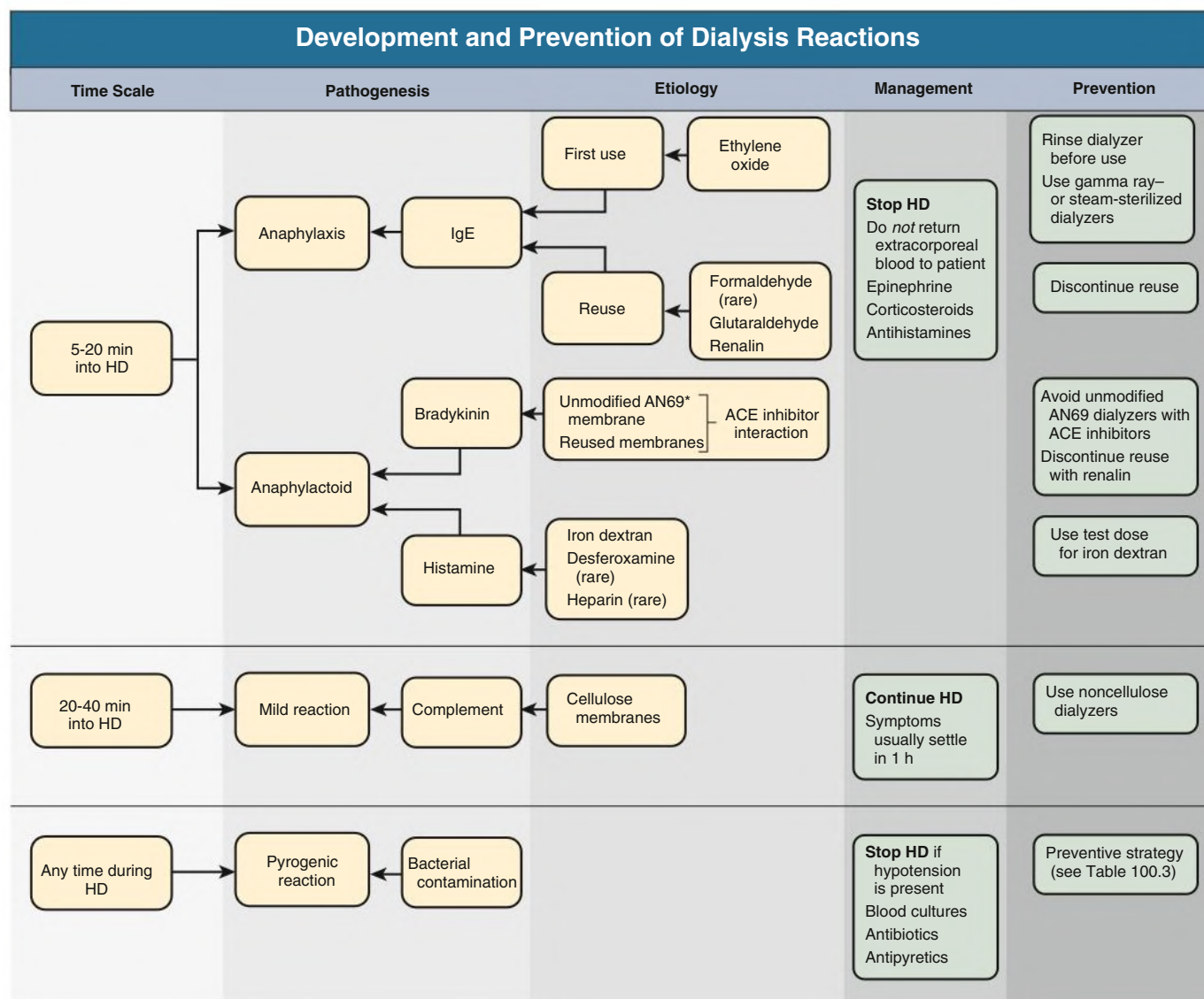


Fig. 100.5 Development and prevention of dialysis reactions. *Acrylonitrile and sodium methallyl sulfonate copolymer. ACE, Angiotensin-converting enzyme; AN69, acrylonitrile; HD, hemodialysis; IgE, immunoglobulin E.

although a delay of up to 20 minutes is possible. Symptoms vary from subtle to severe and include burning or heat throughout the body or at the access site; dyspnea, chest tightness, and angioedema or laryngeal edema; paresthesias involving the fingers, toes, lips, or tongue; rhinorrhea, lacrimation, sneezing, or coughing; skin flushing; pruritus; nausea or vomiting; abdominal cramps; and diarrhea. Predisposing factors include a history of atopy, elevated total serum IgE, eosinophilia, and the use of ACE inhibitors. The etiology of dialysis reactions is diverse, and a thorough investigation is required.

First-Use Reactions

The majority of first-use reactions were ascribed to the manufacturer's dialyzer sterilant ethylene oxide (ETO), which is now rarely used. The potting compound that anchors the hollow fibers in the dialyzer housing acts as a reservoir for ETO and may impede its washout from the dialyzer, leading to sensitization. When it is conjugated to human serum albumin (HSA), ETO acts as an allergen. By use of a radioallergosorbent test (RAST), specific IgE antibodies against ETO-HSA are detected in two-thirds of patients with such reactions. However, 10% of patients with no history of dialysis reactions have a positive RAST result.

Reuse Reactions

Because most residual ETO is washed out of the dialyzer during first use, reuse reactions are likely to be a result of the disinfectants used for dialyzer reprocessing. These agents include formaldehyde, glutaraldehyde, and peracetic acid–hydrogen peroxide in allergic patients; specific IgE antibodies against formaldehyde are occasionally detectable.

Bradykinin-Mediated Reactions

In the early 1990s, anaphylactoid reactions appeared in Europe among patients dialyzed with modified acrylonitrile (AN69) dialyzers who were also taking ACE inhibitors. Investigation of these incidents revealed that binding of factor XII to this sulfonate-containing, negatively charged membrane resulted in the formation of kallikrein and release of bradykinin, which in turn led to the production of prostaglandin and histamine, with subsequent vasodilation and increased vascular permeability. ACE inactivates bradykinin, and therefore ACE inhibitors can prolong the biologic activities of bradykinin.⁴⁹ These membranes have since been chemically modified, thereby reducing this risk, but are still in use for plasmapheresis.

Anaphylactoid reactions also have been observed in patients taking ACE inhibitors who were dialyzed with membranes that had been reprocessed with peracetic acid–hydrogen peroxide; these reactions abated once reprocessing was discontinued, despite continued use of ACE inhibitors. It has been speculated that Renalin may oxidize cysteine-containing proteins that are adsorbed on the dialyzer membrane, leading to the formation of cysteine sulfonate and contact activation of factor XII. Bradykinin-mediated reactions are rarely observed during contemporary hemodialysis treatments.

Drug-Induced Reactions

Anaphylactoid reactions to parenteral iron dextran occur in 0.6% to 1% of HD patients. Significantly higher rates of anaphylactoid reactions have been observed among users of higher molecular weight compared with lower molecular weight iron dextran. In vitro, dextran produces a dose-dependent basophil histamine release. The Kidney Disease Improving Global Outcomes Guidelines recommend that resuscitative medication and personnel trained to evaluate and resuscitate anaphylaxis should be available whenever a dose of iron dextran is administered.⁵⁰ Alternative preparations of iron provoke fewer anaphylactoid reactions than iron dextran and are discussed further in Chapter 86.

Hypersensitivity to heparin formulations is rare and usually responds to substitution of beef with pork heparin or vice versa. An alternative is to substitute low molecular weight heparin. A nationwide outbreak in the United States of severe adverse reactions in HD patients was attributed to vials of heparin contaminated with oversulfated chondroitin sulfate.⁵¹

Treatment and Prevention

Treatment of anaphylactic and anaphylactoid reaction requires the immediate cessation of HD without returning the extracorporeal blood to the patient. Epinephrine, antihistamines, corticosteroids, and respiratory support should be provided, if needed. Specific preventive measures include rinsing the dialyzer immediately before first use, substituting ETO with dialyzers sterilized with gamma ray or steam, avoiding unmodified AN69 membranes in patients taking ACE inhibitors, and discontinuing reprocessing procedures in selected patients.

Mild Reactions

Mild reactions particularly occur 20 to 40 minutes after initiation of dialysis, predominantly with first use of unsubstituted cellulosic dialyzers, and consist of chest or back pain. Dialysis can be continued because symptoms usually abate after the first hour, suggesting a relation to the degree of complement activation. Indeed, these reactions decrease with the use of substituted and reprocessed unsubstituted cellulose membranes. Administration of oxygen and analgesics is usually sufficient. Preventive measures include automated cleansing of new dialyzers and use of noncellulose dialyzers.

Fever and Pyrogenic Reactions

Fever during dialysis can be due to infection or excessive microbial contamination of the dialysis apparatus. The latter, known as *pyrogenic reaction*, is usually a diagnosis of exclusion. Several factors during dialysis place patients at risk for exposure to bacterial products, including contaminated water or bicarbonate dialysate, improperly sterilized dialyzers, use of central venous dialysis catheters, and cannulation of infected arteriovenous fistulas or arteriovenous grafts.⁵² Soluble bacterial products such as endotoxin fragments can diffuse across the dialyzer into the blood, resulting in cytokine production and, consequently, pyrogenic reactions. Whereas high-flux dialyzers

have larger pores that may potentially allow larger fragments to cross from the dialysate to the patient, the synthetic high-flux membranes have a thick wall that tends to be very adsorptive for endotoxin fragments, thus mostly preventing this phenomenon.⁵³ Similarly, although HDF potentially has a higher risk for pyrogenic reactions because of the infusion of 10 to 25 L of dialysate, the customary use of two serial ultrafilters in the dialysate circuit make this an uncommon occurrence. Strategies for the prevention of pyrogenic reactions are summarized in Table 100.3.⁵⁴

When fever develops during HD, the first step is to address hemodynamic stability. If the patient is hypotensive, administration of fluids, cessation of ultrafiltration, and discontinuation of dialysis are often required, and refractory hypotension suggesting severe sepsis should trigger hospitalization.

TABLE 100.3 Strategies to Prevent Bacterial Contamination

Fluid Type	Microbial Count (cfu/mL)	Endotoxin (EU)
Water products	<100	<0.25
Dialysate	<100	<0.50
Ultrapure dialysate (e.g., for HDF)	<0.1	<0.03
Reprocessed dialyzers	No growth	—
Use appropriate germicide:		
• 4% formaldehyde^a • 1% formaldehyde heated to 40°C^{a,b} • Glutaraldehyde^b • Hydrogen peroxide–peracetic acid mixture^b • Heat sterilization (105°C for 20 hours) for reprocessing of polysulfone membranes^b		
Wash and rinse the vascular access arm with soap and water.		
Before cannulation, inspect vascular access for local signs of inflammation.		
Scrub the skin with povidone-iodine or chlorhexidine and allow to dry for 5 minutes before cannulation.		
Record temperature before and after dialysis.		
When central delivery system is used:		
• Clean and disinfect connecting pipes regularly. • Remove residual bacteria or endotoxin by additional filtration.		
When single-patient proportioning dialysis machine is used:		
• Freshly prepare bicarbonate dialysate on a daily basis. • Discard unused solutions at the end of each day. • Rinse and disinfect containers with fluids that meet AAMI standards. • Air dry containers before dialysate preparation.		
Follow manufacturer's guidelines for use of preservative-free medications.		

Strict adherence to AAMI/ISO 11663:2014 standards.

^aA minimum of 11- or 24-hour exposure to peracetic acid or formaldehyde is required, respectively.

^bThese germicides are equivalent or superior to 4% formaldehyde. The action level for the total viable microbial count in the product water and conventional dialysate is 50 cfu/mL, and the action level for the endotoxin concentration is 50% of the relevant standard.

AAMI, Association for the Advancement of Medical Instrumentation; cfu, colony-forming units; EU, endotoxin units; HDF, hemodiafiltration; ISO, International Organization for Standardization.

From Lata C, Girard L, Parkins M, James MT. Catheter-related blood stream infection in end-stage kidney disease: a Canadian narrative review. *Can J Kidney Health Dis.* 2016;3:24.

The next step is to identify a potential source of infection. The dialysis vascular access should be carefully examined. If an infectious source related to non-vascular access is identified, specific therapy should be instituted on the basis of the working diagnosis. Nontunneled and tunneled central venous dialysis catheters always should be suspected as a likely cause of infection, even in the absence of local signs of infection such as erythema and exit site drainage. Catheters with evident signs of infection at the insertion site should be removed and the tip cultured (detailed management is outlined in [Chapter 96](#)).

Antipyretics should be administered, and blood culture specimens should be obtained before initiation of antibiotic therapy; this should include cultures from temporary access devices. The initial choice of antibiotics should include gram-positive and gram-negative bacterial coverage, and the regimen should be adjusted according to the culture results and local guidelines.

In the presence of a dialysis catheter, paired blood culture specimens should be obtained from a peripheral vein and the catheter lumen, and a broad-spectrum antibiotic regimen should be initiated. Catheter lock solutions that use either antibiotics or sterilants such as calcium citrate have been demonstrated to significantly reduce the rate of catheter-related bacteremia.^{55,56} In the case of catheter-related bacteremia, removal of the dialysis catheter is strongly indicated, as is transesophageal echocardiography to rule out endocarditis, particularly with staphylococcal sepsis. A regimen of at least 14 days of antibiotic therapy after removal of the catheter is also recommended.

An outbreak of bacteremia among several dialysis patients involving a similar organism should prompt a thorough search for bacterial contaminants in the dialysis equipment.⁵² Attention should also be paid to multiuse vials that are punctured several times, such as vials of epoetin, which has been linked to an outbreak of bloodstream infection. Single-use vials are preferred, when available.

Investigation of a Dialysis Pyrogenic Outbreak

Although causes of dialysis outbreaks are usually easily identifiable, often the reason for the outbreak is less clear, such as water contamination with bacterial toxins,⁵⁷ medication chemical impurities,⁵¹ bacterial contaminants, systemic embolization of degraded dialyzer membrane polymer after prolonged or improper storage,⁵⁸ and hemolysis from faulty blood tube sets.⁵⁹ Investigation of a dialysis outbreak requires a methodical approach, including a critical review of the medical records and the various steps of the dialysis procedure ([Box 100.1](#)).

MISCELLANEOUS COMPLICATIONS

Postdialysis Fatigue

An ill-defined “washed-out” feeling or malaise during or after HD is a common nonspecific symptom that is observed in about one-third of patients and has multifactorial origins. Reduced cardiac output, peripheral vascular disease, depression, poor conditioning, postdialysis hypotension, hypokalemia or hypoglycemia, mild uremic encephalopathy, myopathy caused by carnitine deficiency, and dialysis-induced cytokine generation have all been incriminated. The use of bicarbonate dialysate with or without additional glucose (5–10 mmol/L) and L-carnitine supplementation (20 mg/kg/day) have been shown to improve postdialysis well-being. A trial of thrice-weekly L-carnitine at 20 mg/kg for 6 months resulted in a marked decrease in C-reactive protein level, which was paralleled by an increase in body mass index.⁶⁰ To date, however, there is no conclusive evidence that L-carnitine improves quality of life in unselected dialysis patients.⁵⁰

Compared with thrice-weekly HD, more frequent dialysis, including short daily and nocturnal HD, has been associated with a marked shortening in the time it takes for patients to recover from a dialysis

BOX 100.1 Investigation of Dialysis Pyrogenic Outbreak

Review of Medical Records

Demographics
Underlying diseases
Dialysis schedule
Dialysis machine
Dialyzer used

- Membrane
- Type
- Manufacturer's sterilization method
- Reuse germicide (if applicable)

Medication history
Signs and symptoms of illness
Laboratory tests
Interview health care workers who were caring for patient during incident

Procedural Review

Water treatment systems and practices

- Disinfection
- Distribution
- Storage procedures

Disinfection and maintenance of reprocessed dialyzers
Disinfection and maintenance of dialysis machines
Review of patient's dialysis sessions

session and to resume their daily activities, which is a surrogate for postdialysis fatigue.

Pruritus

Pruritus is common and can be a very troubling symptom. The cause is often multifactorial, including xerosis, hyperparathyroidism, neuropathy, derangements in the immune system, and inadequate dialysis. In many patients, pruritus is more severe during or after dialysis and may be a manifestation of an allergic reaction to heparin, ETO, formaldehyde, acetate, or the dialysis membrane. In this subgroup of patients, use of gamma ray-sterilized dialyzers, discontinuation of formaldehyde use, switching to bicarbonate dialysate, use of low-dialysate calcium and magnesium might result in cessation of itching. Difelikefalin, a peripherally restricted and selective agonist of kappa opioid receptors, has recently shown significant promise in the treatment of uremic pruritus and depending on availability should be considered.⁶¹ A detailed discussion of the management of uremic pruritus can be found in [Chapter 91](#) (see [Fig. 91.3](#)).

Finally, eczematous reactions to antiseptic solutions, rubber gloves or puncture needle components, puncture needles, or adhesive tapes used to secure dialysis needles also should be considered.⁶²

Genitourinary Problems

Priapism occurs in less than 0.5% of male HD patients. It is not related to sexual activity and occurs while the patient is on dialysis. The patient is usually awakened from sleep by a painful erection. Although the majority of cases are idiopathic, secondary causes include hyperviscosity; high hematocrit from androgen or epoetin therapy; dialysis-induced hypoxemia and hypovolemia from excessive ultrafiltration, particularly in men with sickle cell disease; and use of α -blockers, such as prazosin, or an antidepressant, such as trazodone.⁶³

Urgent urologic referral is mandatory. Acute treatment consists of corporal aspiration and irrigation. Although surgical bypass

provides venous egress from the corpora cavernosa, secondary impotence commonly develops but may be effectively treated by a penile prosthesis.

An unusual case of testicular angina occurring during excessive ultrafiltration has been described. The patient was diabetic and had extensive vascular calcification evident on imaging. He eventually required orchidectomy.⁶⁴ A similar mechanism for this type of “angina” also has been described for the mesentery and right colon.

Hearing and Visual Loss

Intradialytic hearing loss may be caused by bleeding in the inner ear as a consequence of anticoagulation, cochlear hair cell injury from edema, or flux of fluid and electrolytes across cochlear cell membranes.

Intradialytic visual loss is rare but can be caused by central retinal vein occlusion, retinal hemorrhage secondary to heparin exposure in diabetics, precipitation of acute glaucoma, ischemic optic neuropathy secondary to hypotension, or Purtscher-like retinopathy secondary to leukocyte embolization.

SELF-ASSESSMENT QUESTIONS

1. The most common cause of intradialytic hypotension is:
 - A. new-onset sepsis.
 - B. cardiac ischemia.
 - C. intravascular volume depletion.
 - D. altered thermal balance.
 - E. use of antihypertensive medications.
2. The most appropriate preventive strategy to avoid intradialytic hypotension is:
 - A. isothermic dialysis.
 - B. sodium modeling.
 - C. hematocrit-controlled volume removal.
 - D. avoidance of excessive interdialytic weight gain.
 - E. use of agents such as midodrine.
3. Intradialytic hemolysis may be caused by all of the following *except*:
 - A. copper in the dialysate.
 - B. residual formaldehyde.
 - C. chloramine spillover.
 - D. faulty dialysis pumps.
 - E. heparin exposure.
4. Severe dialyzer reactions may occur as a result of exposure to:
 - A. new synthetic dialysis membranes.
 - B. potting compound.
 - C. sterilant effects related to steam or gamma radiation.
 - D. angiotensin-converting enzyme inhibitors.
 - E. peracetic acid.
5. Management options for intradialytic cramps include:
 - A. intravenous hypertonic saline.
 - B. oral magnesium salts.
 - C. oral carnitine supplementation.
 - D. intravenous 50% dextrose.
 - E. all the above.

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